

System Selection & Problem Statement Formulation

1. Target System Selection

- a. Microsoft Admin Portal
- b. The Microsoft admin portal is a jumbled mess of information designed as a key component for system administrators to interact with user accounts. Due to several separate admin centers being created since the initial introduction of the admin portal, the original admin portal has become a kind of obsolete but necessary part of system administrator duties. It suffers from hidden information that used to be a part of the app, but have since been moved to separate admin portals.

2. Interviews

For this and following interviews, I interviewed Nick. He is a former colleague of mine, and a current system administrator.

- a. **Interview 1 – Task: User has MFA, but lost phone and doesn't remember password.**
 - While Nick was able to navigate through this task quite adeptly, he had a few gripe points that he wanted to take note of. The Office 365 admin page is meant to act as a hub, and central point for managing everything to do with an end user. That being said; there is no mention of which admin centers you are actually supposed to use for each task since there are now a dozen different ones. While you can accomplish this task in the Entra admin center, you can sort of accomplish half of it in the Office admin center. You can reset the user's password in the Office admin center, but any attempt to change MFA will take you to the Entra admin center. When moving to the Entra admin center, it does not keep track of which user you were working on, so you have to find them again in the list of every user. See Figure 2 for reference.
- b. **Interview 2 – Task: User left company, but company still needs access to the user's email and one drive.**
 - The email task was pretty straightforward here, but the OneDrive task had some snags. Thankfully, Nick has a fair amount of experience here, so he was able to walk me through the process as well as common complaint points. So the user left the company, but say HR still needs access to that user's old email account. While you could give them read/manage permissions, that wouldn't work when you inevitably end of removing the user's license since it

deletes their mailbox after 30 days. The admin page doesn't warn you about this anywhere. The correct course of actions here is to convert the user's mailbox into a shared mailbox, then create a distribution list, then forward incoming mail from that shared mailbox to the distribution list, then add the HR person to that distribution list. Nowhere on any of the admin pages does it advise you on how to do that or that you need to do that. See figure 6.

- The OneDrive task was simple to solve, but needs a proper way to handle it from the admin side of things. OneDrive has a sort of admin center called SharePoint, but even though it claims to manage OneDrive, it only manages group's OneDrives. To actually give someone else access to another user's OneDrive, you have to first give yourself access, then go into the OneDrive as an admin, then give the second user access, then remove yourself. There should really be a button for that. See figures 1 and 7.

c. Interview 3 – Task: User left the company; the company needs the old user's license provided to a new user.

- This one was probably the most straightforward as far as accomplishing the task is concerned, but most aspects of this task are hidden to the user performing them. There is no warning for the end user that they are going to lose access to their email, it just happens. There is no warning for the admin that by removing their email license, Microsoft will delete all of their stored emails after 30 days. If you want to know what any of the licenses do/did, you have to google them, because there is no description anywhere. If the old user had 1 license, but the new user needs 2 licenses, you may have to purchase a new license. The problem with purchasing a new license is that unless you buy them from Microsoft directly, you will have to contact your cloud reseller and see how much they charge for licenses. There is nowhere to see how much licenses cost unless you contact a third party. Microsoft does this so that cloud resellers can mark up the cost of their licenses. See figures 3, 4, and 5.

Heuristic Evaluation & Accessibility Audit

1. Nielsen's 10 Usability Heuristics Audit

- a. Individual Evaluation & Heuristic Mapping
 - i. **Visibility of System Status**

- Rather than informing the user about when they can expect changes and what the system is doing in the back end. When you assign or remove a license, or update any permission, you are met with a “please allow 24-48 hours for the change to take place” message; even though it often takes only a few minutes. This is frustrating since clients often want live updates for their tickets, but you can’t give them anything to work with.

- **Severity Rating - 2**

ii. **Match Between System & Real World**

- The Office admin portal actually does a decent job with this, using words such as “mailbox” to describe emails, or “OneDrive” to describe the “one drive” that you store everything to. The “Exchange” admin center is where you go to look at mail stuff, because you “exchange” mail. Where this falls off is when they try to add more resources and applications like SharePoint or Perview.

- **Severity Rating - 0**

iii. **User Control & Freedom**

- The user is easily able to leave any menu that they have open, but there is no undo button. The closest to an undo feature would be to look at the audit log, but there are only two of those. One is in Entra, which every admin should have access to, but the other is in Perview, which you need a special license to access.

- **Severity Rating - 3**

iv. **Consistency & Standards**

- Even though the main page is called the “Microsoft 365 Admin Center”, there are actually 12 total admin centers. The 365 admin center was supposed to act as the hub where you could do everything, but a combination of retired, but still used, admin centers, as well as new admin centers for features that weren’t yet created when the 365 admin portal was created, means that the 365 admin center is really just a gathering of half-working features. They also use the word “devices” to both refer to MFA devices as well as Intune devices, making the 365 admin portal that much more confusing.
- **Severity Rating - 1**

v. Error Prevention

- Since the admin center is designed for system administrators, there is very little error prevention or error checking. You can remove the entire catalogue of licenses from every user without a confirmation, and that could bring everything in the company to a halt.
- **Severity Rating - 4**

vi. Recognition Rather than Recall

- The closest thing to this that the Office admin portal has is that you can pin frequently used resources to the side menu so that you can find them again easily. Because there are so many links to other admin portal, it often doesn't save what user you might have been working on when you move to the new page.
- **Severity Rating - 2**

vii. Flexibility & Efficiency of Use

- There are no keyboard shortcuts, and quite frankly, an inexperienced user is likely to get lost in all of the different admin centers and what they are in charge of. The typical job requires hitting at least 2 different admin centers, and there is no guide on the page to tell you where to go. There is a PowerShell menu that you can access to do most tasks, but rather than being a speedy solution for experienced users, it turns out to be more of a hassle that you have to deal with from time to time because it is necessary for a few things, but unable to do a lot.
- **Severity Rating - 3**

viii. Aesthetic & Minimalist Design

- Because of the redundant admin centers, many of the features on the Office 365 admin page just lead you to their respective admin centers. This means that not only do you have to remember which user you may have been working on, since it doesn't pick up where you left off on this new page, but it also means that there are several redundant ways of getting the same task accomplished. Just in the user page alone there are probably 5+ ways to get to the Entra admin center.
- **Severity Rating - 1**

ix. Help Users Recognize & Recover from Errors

- There is very little in the way of error assistance on the Office admin page. If you go to reset a user password, sometimes the one you have selected will work, sometimes it doesn't. If it doesn't, it won't tell you why, it will only tell you that it doesn't meet the complexity requirements. No where does it mention what those complexity requirements are. You have to go and search the web for the Microsoft password complexity requirements.

- **Severity Rating - 2**

x. **Help & Documentation**

- While there is virtually nothing in the way of assistance inside of the admin portal, there are quite extensive guides on how to do most everything in the Microsoft forums. 9 times out of 10, you will be going to the Microsoft forums for assistance if you have any problems in the Admin centers.

- **Severity Rating - 0**

2. **Baseline Accessibility (WCAG 2.1 AA) Audit**

a. **Visual color contrast ratios**

- i. Just on the user page there are 52 contrast errors.

b. **Keyboard navigation accessibility and visible focus indicators**

- i. Tab targeting makes you go through every top menu as well as every side menu item before you get to the contents of the page. Assuming that I took the time to get to the user page (which you can't even tab target to) it would take you 36 different tab presses before you can actually manage a single user's account.

c. **Semantic document structure and aria-* attribute support for screen reader compatibility**

- i. The page is screen reader compatible.

Post-Mortem Reflection

1. How did combining qualitative user observations with an expert heuristic evaluation reveal usability flaws that a developer testing their own code would likely overlook?
 - I believe that the main problem with the page I am evaluating can be summed up as a "too many cooks in the kitchen" problem. Once upon a time this page was the main way to handle everything related to Microsoft products and licenses, but there have been features both added and removed since then. Every new feature wants its own admin center so that

Microsoft can charge a premium for it. This has caused the original admin center to be a weird mash of 12 different admin centers in one, fighting for its own identity. Since every important menu has its own admin center now though, the original hub has become rather neglected. Developers are likely to overlook the Office 365 admin center because they have their own admin center now, and why would you use the outdated original admin center when you can use the new and shiny one?

2. Identify one heuristic violation that directly creates a legal or ethical accessibility barrier under WCAG 2.1 AA, and explain how resolving that violation improves the user experience for both disabled users and the general public.
 - I think the most glaring heuristic violation was the 52 different contrast errors. Microsoft has gone for a “sleek” white text on a black background look, and that often includes things like graphs. By improving the contrast ratio as well as making some of the texts a bit larger and more legible, I think that the page would be greatly improved for those with visual impairments.

Images:

Name	Description
 Azure	The Azure portal is a unified, web-based interface that enables users to manage, monitor, and configure all their Azure services and resources in one place.
 Exchange	Manage advanced email settings, such as quarantine, encryption, and mail flow rules.
 Microsoft 365 app	Manage, configure, and monitor deployment of Microsoft 365 Apps for your organization.
 Microsoft Entra	Use the Microsoft Entra admin center to manage identities, permissions, and network access.
 Microsoft Purview	Use Microsoft Purview to manage data governance, and risk and compliance solutions for all your data.
 Power Platform	Manage your data analysis with Power Apps, Power Automate, and Dynamics 365.
 Search & intelligence	Manage Microsoft Search settings including services and content that are available for people in your organization. Make finding internal tools, documents, and people just as easy as searching the web in Bing.
 Security	Use Microsoft 365 Defender for unmatched visibility into threats, track assets, and deploy security updates. Respond to incidents, and manage your security posture.
 SharePoint	Manage sites, sharing, storage, and more for SharePoint and OneDrive. Migrate files and sites to Microsoft 365.
 Teams	Set up and manage options for Microsoft Teams chat, meetings, guest access, and more.
 Viva Engage	Manage your Yammer network, set a usage policy, control external network settings, and enable features like translation.

Azure is no longer used, replaced by Entra

These are only available with the proper license, but it doesn't tell you if you can use them here.

This says OneDrive, but it is only used for group pages not individual users

Figure 1: Multiple redundant admin centers which the user may or may not have access to.

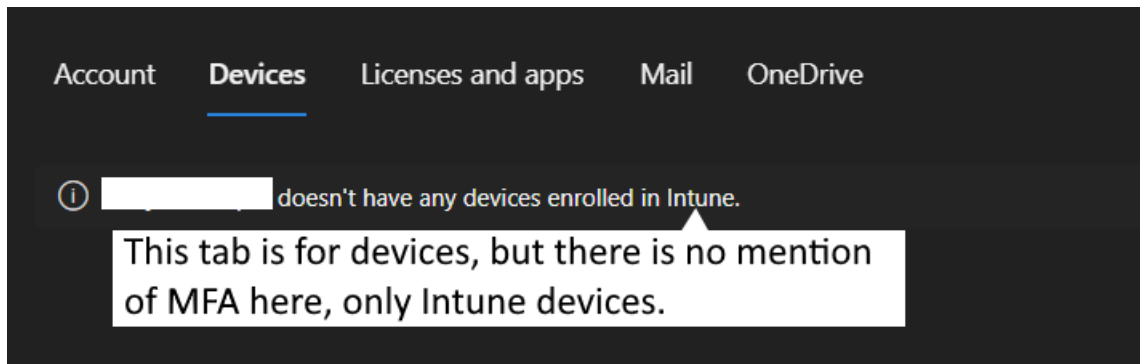


Figure 2: The Device tab of the user page which should also contain MFA devices, but instead only contains Intune devices. Intune not linked in this tab.

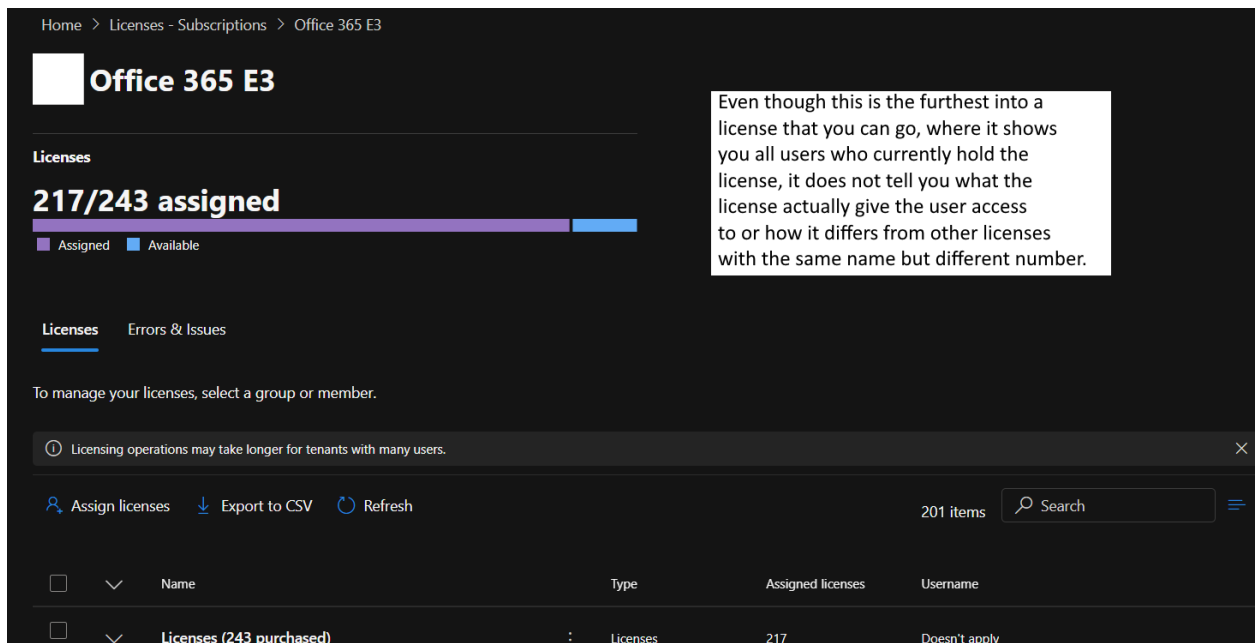


Figure 3. Office 365 E3 license page which does not contain any information on the license properties.

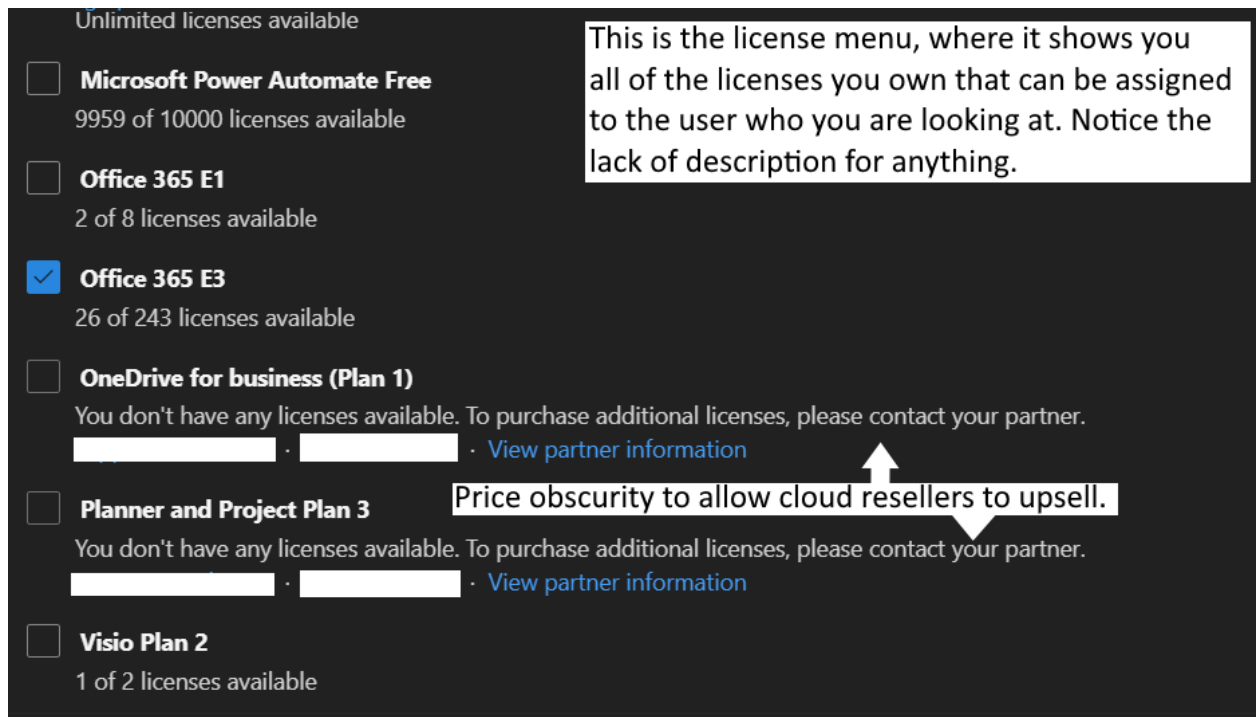


Figure 4: The user tab license page which does not contain license information or the ability to purchase new licenses without reseller intervention.

Name ↑	Available licenses	Assigned licenses	Account type
Dynamics 365 Sales Professional	9999	1/10000	Organization
☐ Exchange Online (Plan 1)	This is a separate menu that shows all available licenses, but no information on what those licenses give access to.	2/4	Organization
☐ Microsoft 365 Audio Conferencing		0/1	Organization
☐ Microsoft Fabric (Free)		2/1000000	Organization
Microsoft Power Automate Free	9959	41/10000	Organization
☐ Office 365 E1	2	6/8	Organization
☐ Office 365 E3	26	217/243	Organization
☐ OneDrive for business (Plan 1)	0	1/1	Organization
☐ Planner and Project Plan 3	0	1/1	Organization
☐ Visio Plan 2	1	1/2	Organization

Figure 5: The license page which also doesn't contain any properties or information.

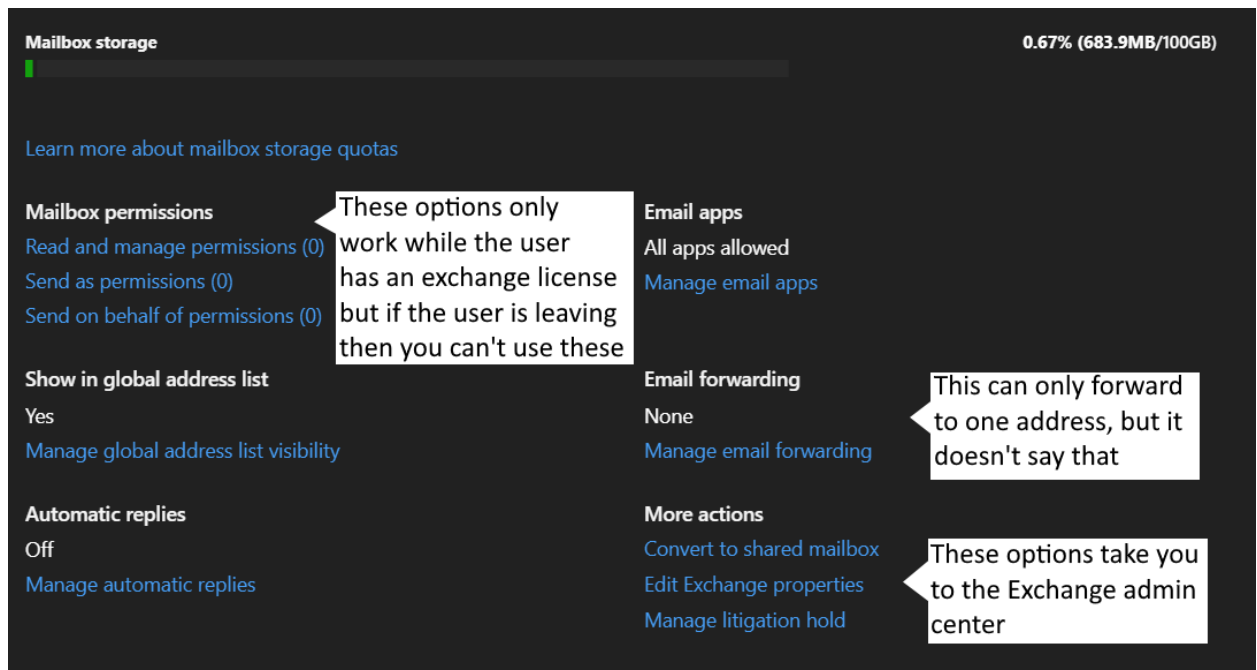


Figure 6: Mail tab of user page. Contains outdated links, and lack of information for how to handle certain procedures.

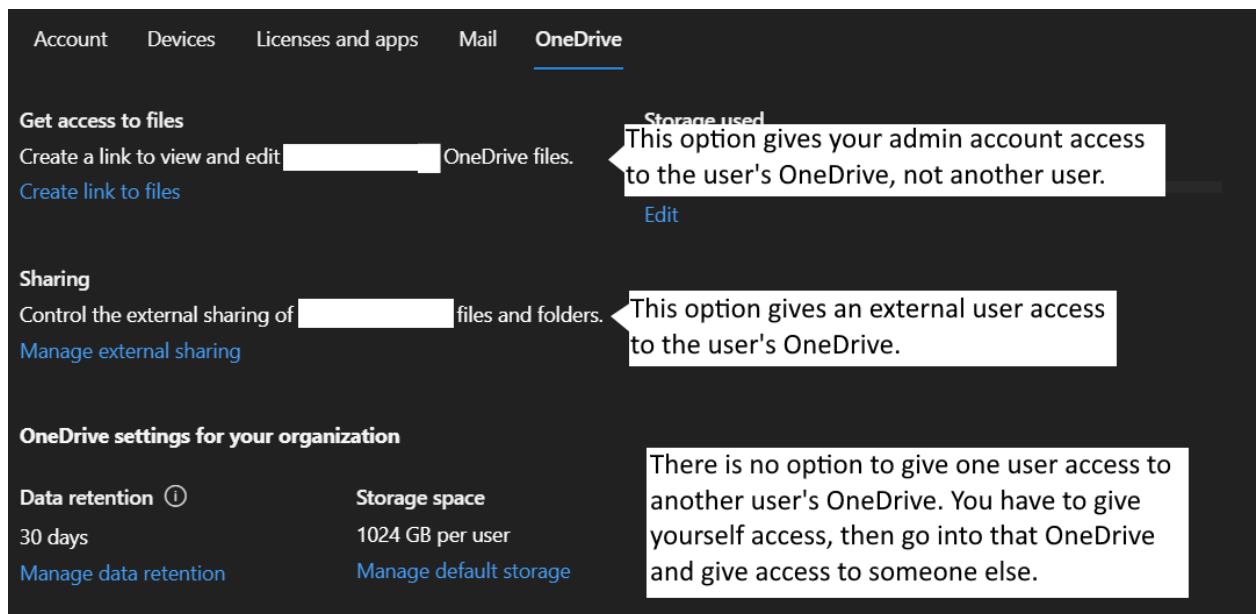


Figure 7: OneDrive tab of user page. Contains some relevant options, but not all options necessary. Does not link to another admin page. The OneDrive admin page does not actually manage individual users' OneDrives.